

Describe the mathematical model used in lossless compression. Explain how probability models and physical models contribute to efficient data representation with egs.

Ans: Mathematical Models used in Lossless Compression

Lossless Compression reduces file size without losing any information. The original data can be perfectly reconstructed. This is achieved using mathematical models that predict or represent data efficiently.

The two main models are:

1. probability (Statistical) Models
2. physical (Source) Models

probability Models

probability model assigns probabilities to symbols based on how frequently they occur. More frequent symbols get shorter codes, and rare symbols get longer codes.

Mathematical Basis

if a symbol has probability $P(x)$, its optimal code length is:

$$L(x) = -\log_2 P(x)$$

This means:

- High probability $\rightarrow$ Small Code length
- Low probability $\rightarrow$ large Code length

eg: Huffman Coding

Huffman Coding builds a binary tree using symbol frequencies.

Example data :

A A A B B C

Symbol	frequency	probability	Code (Huffman)
A	3	0.5	0
B	2	0.33	10
C	1	0.17	11

frequency symbol A gets shortest Code $\rightarrow$ efficient Compression

eg 2: Arithmetic Coding

Arithmetic Coding encodes the entire message as a single fractional number between 0 and 1.

Instead of assigning separate Codes :

- It narrows a range based on probabilities.
- Very close to theoretical entropy limit

used in :

- JPEG
- PNG
- H.264

Advantages of probability Models

- Achieve Compression near entropy limit
- Adapt to data statistics
- work well for text, images, and general data.

Physical Models (Source Models)

Physical models use knowledge about how data is generated. Instead of just symbol frequency, they model:

- patterns
- Dependencies
- Correlation
- Structure

eg: Run-Length Encoding (RLE)

eg: AAAAABBBBBCC

Stored as

(A,5) (B,4) (C,2)

very efficient to:

- Black-white images
- Fax documents

eg2: Lempel-Ziv (Dictionary Model)

Instead of repeating text:

- Replace repeated strings with pointers.

used in:

- Zip
- GIF
- PNG

eg: text = ABABABAB

encoded as = AB (position = 1, length = 6)

eg3: predictive Models

predict next symbol using previous symbols.

if prediction is correct:

only small error is stored

used in

- FLAC
- PNG

Comparison: probability Vs Physical Models

Feature	probability Model	physical Model
Based on	Symbol frequency	data structure & pattern
Mathematical Tool	entropy, $\log_2$	Correlation, prediction
example	Huffman Coding	LZ77
Best for	General text	structured / repetitive data

How they contribute to efficient data Representation

probability Models

- Remove statistical redundancy
- Approach Shannon entropy limit
- Minimize average code length

Physical Models

- Remove structural redundancy
- exploit repetitions & correlation
- Convert data into more compressible form

Lossless Compression use entropy and symbol probabilities to assign optimal code lengths.

- probability models use entropy and symbol probabilities to assign optimal code lengths.
- physical models exploit repetition, structure, and source characteristics to reduce redundancy
- Together, they ensure efficient data representation without information loss.